

Two biogeographic factors in particular—the latitude of a community and the area it occupies—have been investigated for more than a century.

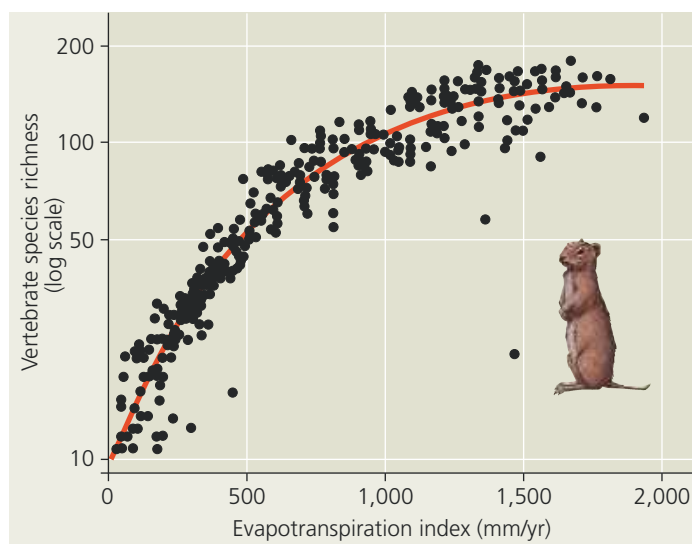
Latitudinal Gradients

In the 1850s, both Charles Darwin and Alfred Wallace pointed out that plant and animal life was generally more abundant and diverse in the tropics than in other parts of the globe. Since that time, many researchers have confirmed this observation. One study found that a 6.6-hectare (1 ha = 10,000 m²) plot in tropical Malaysia contained 711 tree species, while a 2-ha plot of deciduous forest in Michigan typically contained just 10 to 15 tree species. Many groups of animals show similar latitudinal gradients. For instance, there are more than 200 species of ants in Brazil but only 7 in Alaska.

Two key factors that can affect latitudinal gradients of species richness are evolutionary history and climate. Over the course of evolution, a series of speciation events may lead to increased species richness in a community (see Concept 24.2). Tropical communities are generally older than temperate or polar communities, which have repeatedly “started over” after major disturbances such as glaciations. As a result, species diversity may be highest in the tropics simply because there has been more time for speciation to occur in tropical communities than in temperate or polar communities.

Climate is another key factor thought to affect latitudinal gradients of richness and diversity. In terrestrial communities, the two main climatic factors correlated with diversity are sunlight and precipitation, both of which occur at high levels in the tropics. These factors can be considered together by measuring a community’s rate of **evapotranspiration**, the evaporation of water from soil and plants. Evapotranspiration, a function of solar radiation, temperature, and water availability, is much higher in hot areas with abundant rainfall than in

▼ **Figure 54.29 Energy, water, and species richness.** Vertebrate species richness in North America increases with an index of evapotranspiration, expressed as rainfall equivalents (mm/yr).



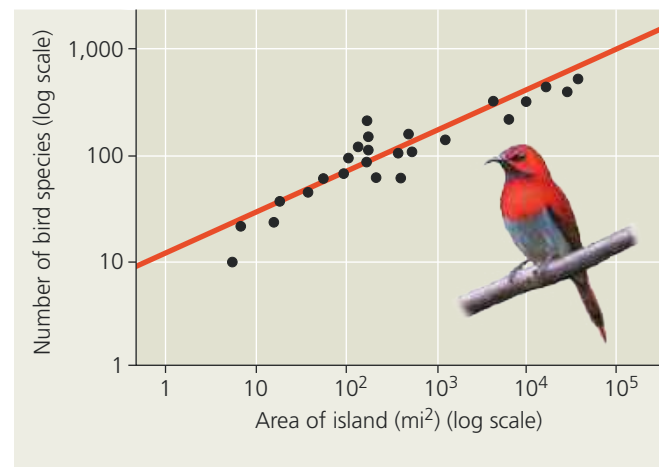
areas with low temperatures or low precipitation. The species richness of plants and animals correlates with measures of evapotranspiration, as shown for vertebrates in **Figure 54.29**.

Area Effects

In 1807, naturalist and explorer Alexander von Humboldt described one of the first patterns of species richness to be recognized, the **species-area curve**: All other factors being equal, the larger the geographic area of a community, the more species it has. One explanation for this relationship is that larger areas offer a greater diversity of habitats and microhabitats.

Predictions of the species-area relationship have been tested by examining the number of animals and plants in many different regions. As one example, in the Sunda Islands of Malaysia, the number of bird species increased with island size (**Figure 54.30**). Different species-area curves vary in how rapidly species richness increases with area. Even so, the basic concept of diversity increasing with increasing area applies in many situations, from surveys of ant diversity in New Guinea to studies of plant species richness on islands of different sizes.

▼ **Figure 54.30 Species richness and island area.** The number of bird species on the Sunda Islands of Malaysia increases with island size.



Island Equilibrium Model

Because of their isolation and limited size, islands provide excellent opportunities for studying how area and other factors affect the species diversity of communities. By “islands,” we mean not only oceanic islands, but also habitat islands on land, such as lakes, mountain peaks, or habitat fragments—any patch surrounded by an environment not suitable for the “island” species.

Recognizing the opportunities that studying islands provides, Robert MacArthur and E. O. Wilson developed a method for predicting the species diversity of islands (**Figure 54.31**). In their approach, the number of species on an island represents a balance between the immigration of new species to the island and the extinction of species already there.

In Figure 54.31, note that the immigration rate *decreases* as the number of species on the island gets larger, while the